Machine learning is here to help

If machine learning and artificial intelligence weren't already doing enough to amaze us, a recent finding published in a research study led by an undergrad student at the University of Toronto has suggested that machine learning could potentially help us find extraterrestrial life. Once every few years talks about the existence, and the chance of us contacting them takes the scientific community by storm. And, if the suggested theories are anything to go by, then we might see another wave of commentary coming our way about the possibilities of establishing contact or just finding extraterrestrial life.

A team of 17 researchers, including Ma, published a research paper titled "A deep-learning search for technosignatures from 820 nearby stars" in the journal - Nature Astronomy. There, the team has theorised about a machine learning program that would be able to process data sets that contain petabytes of data. Classical techniques that are used till now are not capable of handling such large data sets.

> "These results dramatically illustrate the power of applying modern machine learning and computer vision methods to data challenges in astronomy, resulting in both new detections and higher performance. Application of these techniques at scale will be transformational for radio technosignature science."
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> **Cherry Ng,**
> Research advisor and astronomer at SETI Institute and the French National Center for Scientific Research

When working on this research project, the team looked through a data set containing 150 TB of data of over 800 stars. This data set had been previously looked through in 2017 using older techniques. During their re-run, they came across several interesting findings, including radio signals that had not been detected in the first place in the 2017 analysis of the same data set. These signals, as published in the materials by the SETI institute, had the following characteristics:

1. The signals were narrow band, meaning they had narrow spectral width, on the order of just a few Hz. Signals caused by natural phenomena tend to be broadband.
2. The signals had non-zero drift rates, which means the signals had a slope. Such slopes could indicate a signal's origin had some relative acceleration with our receivers, hence not local to the radio observatory.
3. The signals appeared in ON-source observations and not in OFF-source observations. If a signal originates from a specific celestial source, it appears when we point our telescope toward the target and disappears when we look away. Human radio interference usually occurs in ON and OFF observations due to the source being close by.

As of now, as per the researchers, there are two very basic yet important applications of this new ML program and algorithm. The first is that this program will enable a quicker and more effective analysis of data that is collected as a part of various research programs. And if there is a need, then the researchers will be able to re-run their analysis in much shorter periods and potentially discover new information.

Link to the research paper: https://dgit.in/spacemlai